From Vision-Language-Action to Motion:
Mapless LiDAR Occupancy-Guided Trajectory Refinement

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Motivation

Indoor robots usually navigate on a prebuilt map, which takes a survey to build and dates as soon as furniture moves. A **vision-language-action (VLA) policy** needs no map: from one camera image and a standing instruction it generates a navigation trajectory directly. But it is trained on clean demonstrations and has no representation of free space, so three things go wrong:

- **Turning.** The policy undershoots, so the path cuts into the inside corner of the junction.
- **Obstacles.** A chair or bin in the corridor is not part of what the policy learned, so the path runs through it.
- **Other people.** Someone walking toward the robot is a moving constraint the policy cannot react to.

We check the trajectory against a **single-scan LiDAR occupancy field** and correct it where it is unsafe, on every scan. The instruction chooses the route; occupancy keeps the motion safe.

Objectives

1. **Map-free policy:** turn one camera frame and a standing instruction into a waypoint trajectory, on the robot itself.
2. **Single-scan geometry:** read centring, openings and junction type from one LiDAR scan, with nothing stored between frames.
3. **Stable tracking:** follow the trajectory with pure pursuit, with LiDAR taking lateral authority in corridors.
4. **Obstacle clearance:** slow and stop for objects that are not part of what the policy learned.

System Overview

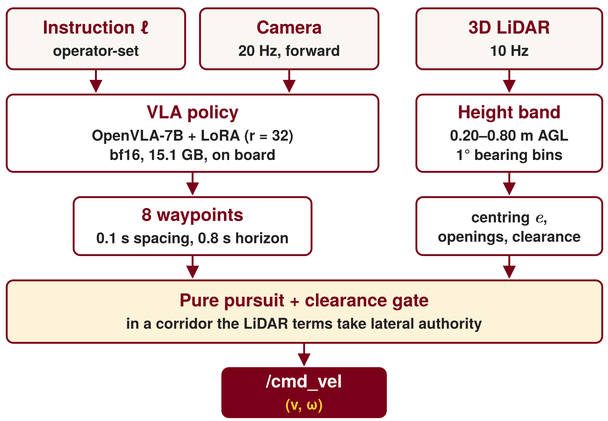


Fig. 1. Two tracks meet at the controller, and nothing leaves the robot.

1 On-Board Policy

One RGB frame and a standing instruction ℓ map to a short waypoint trajectory [3] in the robot's body frame:

$$\pi_\theta : (I_t,\ \ell) \rightarrow (p_1,\ \ldots,\ p_8) \in \mathbb{R}^{16}$$

The points are cumulative positions over 0.8 s, not per-step deltas, so a turn accumulates into an endpoint offset an L_1 objective cannot ignore.

LoRA adapts the backbone at rank 32 — ≈29M of 7.5B parameters — with the vision encoder frozen [1, 2].

On the robot it occupies 15.1 GB in bf16, unquantized. One prediction is a single forward pass — the head emits all eight waypoints at once, not token by token.

2 Single-Scan LiDAR Geometry

One scan, no map, no accumulation. Points are kept in a band at robot height, then reduced to one minimum range per 1° bearing bin.



Fig. 2. Recycling bin beside the robot.

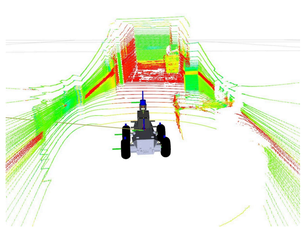


Fig. 3. Robot and its LiDAR point cloud.

Centring. With d_L, d_R the median side-sector ranges,

$$e = (\ d_\mathrm{R} - d_\mathrm{L}\ ) \ / \ 2$$

is the offset from the centreline, positive to the left — meaningful only between two walls, so it is held off at junctions.

Openings. A run of bins beyond the open range counts as a passage only if it is wide enough for the robot at its depth. Runs split at an interior dip deep relative to its flanking peaks, so a wall between two branches separates them.

Labels. Each opening is named by bearing — forward within ±25°, left or right beyond it — and the scene is called a corridor, a branch, a T-junction or a dead end only once the same label holds for five consecutive scans.

Band. The band is measured from the *floor*, not from the sensor: $base_link$ rides 0.66 m above the ground, so a band set in raw sensor height would sample chest level and miss every low obstacle. The floor plane is recovered from the scan itself. Anything outside the band — a cable on the floor, an overhang — stays invisible.

No memory. Nothing carries between scans, so an object that moves is gone on the next one and there is no stale map to clear. The cost: a branch stays occluded behind a corner until the robot is a metre or two away, so LiDAR confirms a junction rather than announcing one.

3 Control and Safety

Pure pursuit. The first waypoint past the lookahead distance fixes the unique arc through the origin tangent to the heading [4]:

$$\kappa = \frac{2 y_\mathrm{L}}{{x_\mathrm{L}}^2 + {y_\mathrm{L}}^2}, \qquad \omega = v\ \kappa$$

Authority, not summation. Summing a centring term onto the policy's steering makes two controllers argue and the robot weave. In a corridor the LiDAR terms own ω, including a heading term — lateral error alone cannot correct a base *pointed* at a wall.

Clearance gate. Clearance scales speed from 1.80 m to zero at 0.70 m; inside 0.40 m the stop is unconditional and overrides the policy [5].

Clear	Slow	Stop
≥ 1.80 m full speed	1.80 → 0.70 m scales to zero	< 0.40 m overrides policy

4 Obstacle Avoidance

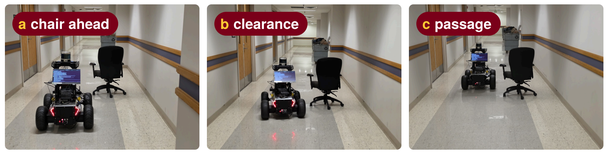


Fig. 4. (a) A chair partially blocks the corridor; the robot takes the open space on the left. (b) LiDAR gives the local geometry of the gap for clearance-aware control. (c) Alongside the chair, with the corridor open ahead.

Decision vs. Timing

A standing commitment, not a step command. Told to turn well before a held-out junction, the policy keeps going straight.

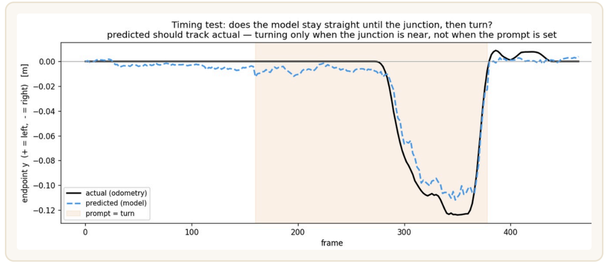


Fig. 5. Endpoint lateral offset, predicted vs. odometry. Shaded: turn instruction active.

1. **Hold.** The endpoint stays near zero well after the instruction is set, while the corridor ahead is unchanged.
2. **Turn.** It deflects only once the junction enters view, tracking the driver's own turn.
3. **Return.** It comes back to zero as the turn completes, and corridor following resumes.

Platform



Base	AgileX Hunter 2.0, Ackermann, ≈1.7 m turning radius
Sensing	3D LiDAR at 10 Hz; one forward RGB camera
Compute	Jetson AGX Orin 64 GB on board, ROS 2 Humble
Rates	perception 10 Hz · control 20 Hz

Conclusion

A 7B policy runs on the robot itself, turning one frame and a standing instruction into a trajectory that single-scan LiDAR keeps centred and clear, with no map in the loop. Timing is learned from the image, so an instruction can be issued well ahead of the junction and need not be timed by whatever supplies it.

Future Work

- **Closed-loop evaluation:** success rate, minimum clearance and interventions over repeated runs, compared with the VLA alone.
- **People in motion:** track people across scans so the robot can anticipate someone walking toward it.
- **Recovery data:** off-centre and human-corrected driving, so the policy learns to recover, not only to follow clean demonstrations.
- **Fuller turns:** close the remaining gap between the predicted turn and the driven one.
- **More buildings:** new layouts and junction types beyond the one building tested so far.

Acknowledgements

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